

Week 11/12 Completion Status:

- Drafted Part I of the research manuscript: Abstract, Introduction, Related Work, and Methodology
- Successfully integrated these sections into the IEEE conference template using Overleaf.
- Enforced mathematical rigor by formally defining the Markov Decision Process (MDP) and Constrained Optimization equations in the Methodology section.
- Updated the first set of figures to high-quality .png formats and matched the citation links to my Annotated Bibliography.
- Git link: <https://github.com/scr200/CP395-Research-Project>
- **Manuscript (overleaf link):**
<https://www.overleaf.com/4338359642wmpyxgbxcrnk#0356be>

Methodology/Changes:

This weekly report focuses on the academic writing and formatting the first half of my paper. I took the raw experimental data from my early EDA and baseline testing and converted it into a publication-quality narrative.

A major change I made was heavily expanding the "Methodology" section. I realized that to meet top-tier venue standards, I couldn't just explain the code; I had to explain the math behind it. I went back to my Week 5 and Week 6 notes and explicitly wrote out the Cost Function and the Scaling Rate constraints. I also made sure to include the exact metrics that justified my design choices, like the "Bursty" workload having a peak-to-mean ratio of 6.10, and the ARIMA baseline failing with a high MAE of 131.86. This proves my autoscaler is grounded in realistic physics, not just theoretical machine learning.

Issues/Risks/Mitigations:

- **Mitigation:** I went through the \bibitem LaTeX code block at the bottom of the document and manually updated every reference to perfectly match the IEEE standard (e.g., abbreviating first names, italicizing journal titles).
- **Risk:** The manuscript was getting too dense with technical jargon, which risked burying the actual business value of the system.
- **Mitigation:** I rewrote the Introduction to focus heavily on the "Reaction Gap" and the financial dangers of AI hallucinations, ensuring the story of the paper remained clear and engaging.